ADVANCED
IOT SMART ROOM
AUTOMATION SYSTEM

2026
PROJECT REPORT

ESP32 • MQTT • NODE-RED • FIREBASE REALTIME DATABASE

MONITOR • AUTOMATE • ANALYZE • CONTROL

PREPARED BY

Ahmad Azroun

Renewable Energy Manager

IoT & Smart Energy Systems Developer

- Real-time Monitoring
- Smart Automation
- Cloud Connected
- MQTT Communication
- Data Visualization

SCAN ME
GitHub Repository

ESP32 Microcontroller	WIRELESS Wi-Fi Connectivity	SENSORS Temp • Hum • Light Motion Detection	MQTT Publish / Subscribe Messaging	NODE-RED Dashboard & Flow Automation	FIREBASE Realtime Database Cloud Storage	DASHBOARD Real-time Charts & Controls	ACTUATORS Fan Control RGB LED

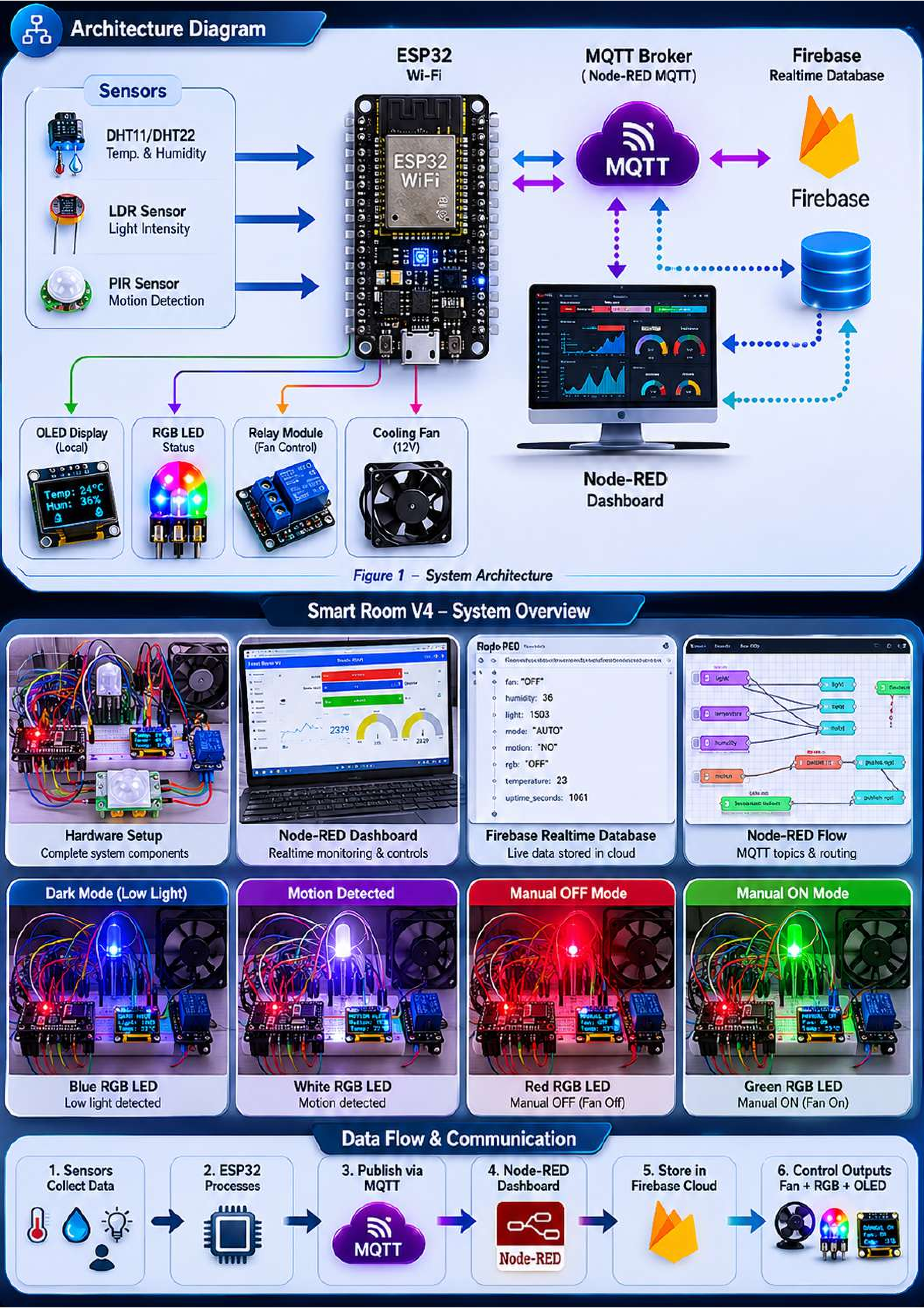


Figure 1 – System Architecture

SMART ROOM V4

ADVANCED IoT SMART ROOM AUTOMATION SYSTEM

ESP32 | MQTT | Node-RED | Firebase

Prepared by: Ahmad Azroun
Renewable Energy Manager | IoT & Smart Energy Systems Developer

DEUTSCHE DOKUMENTATION

Projektübersicht:

Smart Room V4 ist ein professionelles IoT-Automatisierungssystem basierend auf ESP32, MQTT, Node-RED und Firebase Realtime Database. Das System wurde entwickelt, um Raumdaten intelligent zu überwachen und automatisch auf Umgebungsänderungen zu reagieren.

Projektfunktionen:

- Temperatur- und Luftfeuchtigkeitsüberwachung
- Bewegungserkennung mit PIR-Sensor
- Lichtintensitätsmessung mit LDR
- Automatische Lüftersteuerung
- RGB-LED Statusanzeigen
- Echtzeit-Dashboard mit Node-RED
- Cloud-Datenspeicherung über Firebase
- MQTT-basierte Kommunikation
- Manueller und automatischer Betriebsmodus

Arbeitsprinzip:

Die Sensoren senden Daten an den ESP32. Anschließend verarbeitet der ESP32 die Werte und sendet die Daten über MQTT an Node-RED und Firebase. Das System steuert automatisch Lüfter und RGB-Beleuchtung basierend auf Temperatur, Licht und Bewegungserkennung.

ENGLISH DOCUMENTATION

Project Overview:

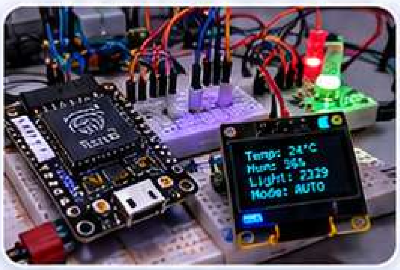
Smart Room V4 is a professional IoT automation system based on ESP32, MQTT, Node-RED, and Firebase Realtime Database. The system was designed to intelligently monitor room conditions and automatically react to environmental changes.

Main Features:

- Temperature and humidity monitoring
- PIR motion detection
- Light intensity measurement using LDR
- Automatic fan control system
- RGB LED smart status indication
- Real-time dashboard using Node-RED
- Cloud storage using Firebase
- MQTT-based communication
- Manual and automatic operating modes

Working Principle:

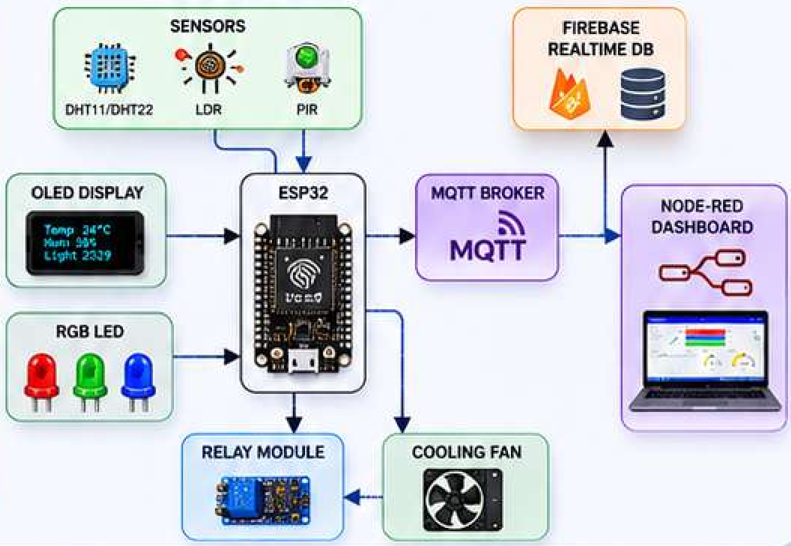
Sensors continuously send realtime data to the ESP32 controller. The ESP32 processes the values and publishes them through MQTT to Node-RED and Firebase. The system automatically controls the cooling fan and RGB lighting based on motion, temperature, and room light conditions.



COMPONENTS USED

Component	Purpose
ESP32	Main controller and Wi-Fi communication
DHT11/DHT22	Temperature and humidity sensing
LDR Sensor	Light intensity monitoring
PIR Sensor	Motion detection
RGB LED	Visual room status indication
Relay Module	Fan switching control
Cooling Fan	Automatic room cooling
OLED Display	Realtime local display
Node-RED	Realtime dashboard and visualization
Firebase	Cloud realtime database storage

ARCHITECTURE DIAGRAM

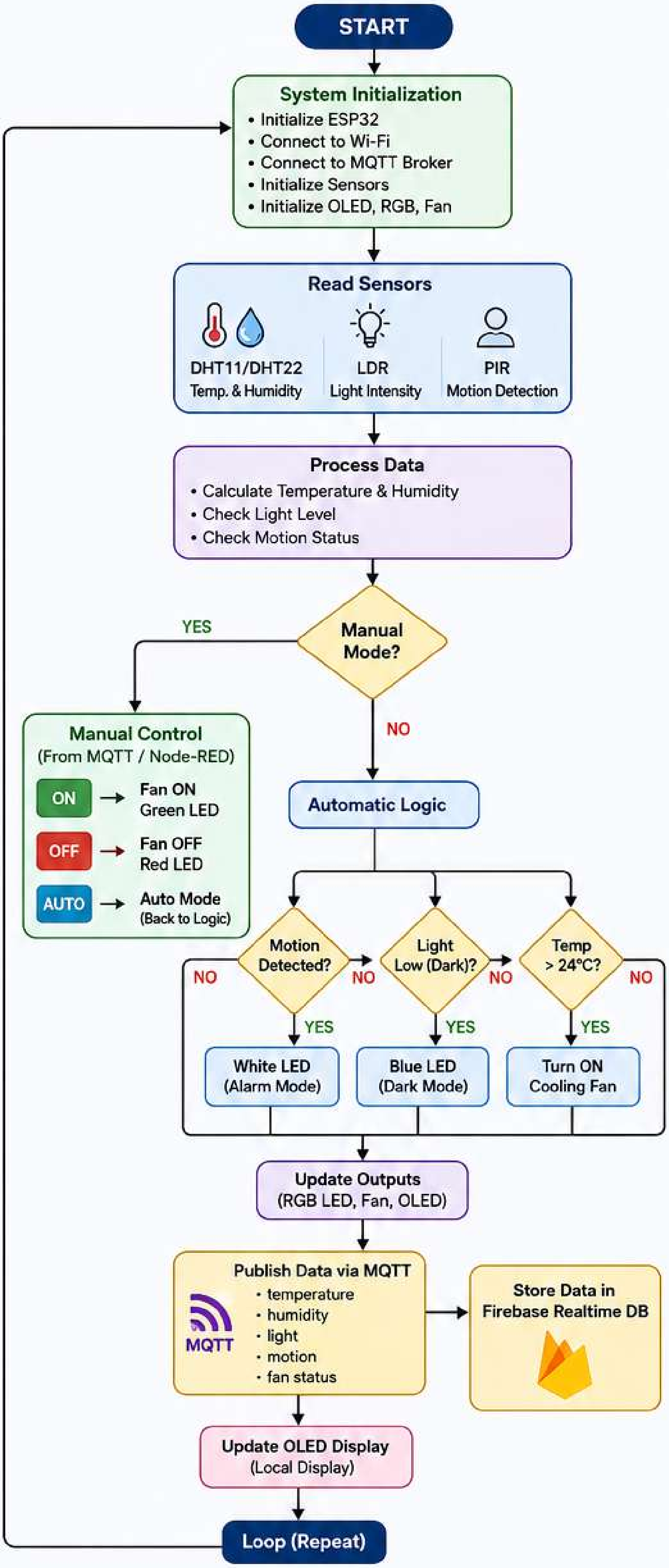


REALTIME MONITORING | SMART AUTOMATION | CLOUD CONNECTED | ENERGY EFFICIENT | SECURE & RELIABLE

SMART ROOM V4 – FLOWCHART

ESP32 IoT Smart Automation System

Sensors • ESP32 • MQTT • Node-RED • Firebase • Actuators • Dashboard

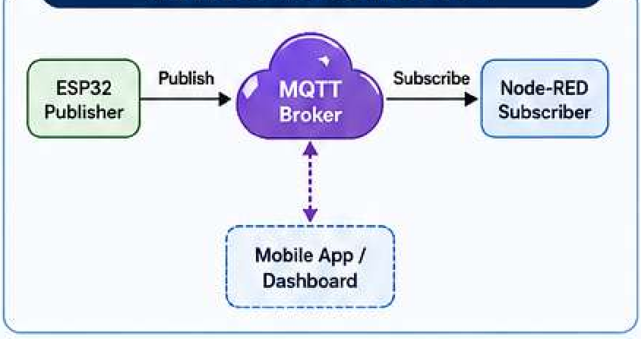


WORKING PRINCIPLE

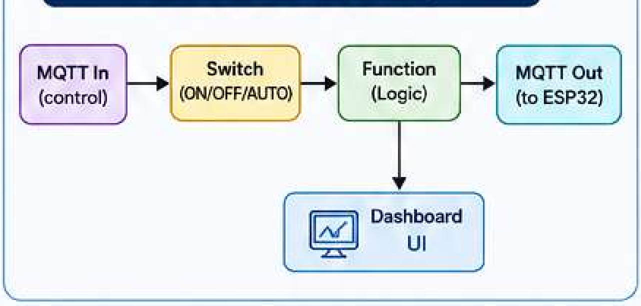
Sensors collect real-time data and send it to ESP32. The ESP32 processes the data and makes decisions based on conditions or user commands (via MQTT). Actions are executed (Fan, RGB, OLED) and data is published to MQTT Broker, visualized on Node-RED Dashboard and stored in Firebase Cloud.

Scan for
Project Demo

MQTT COMMUNICATION FLOW



NODE-RED FLOW (LOGIC)



DATA FLOW TO CLOUD

RGB LED STATUS MODES

Mode / Condition	RGB LED Color	Description
Dark Mode (Light Low)	Blue	Low light environment
Motion Detected	White	Motion alert mode
Manual OFF	Red	System turned OFF
Manual ON	Green	System turned ON
Normal / Idle	Purple	Normal conditions

LEGEND

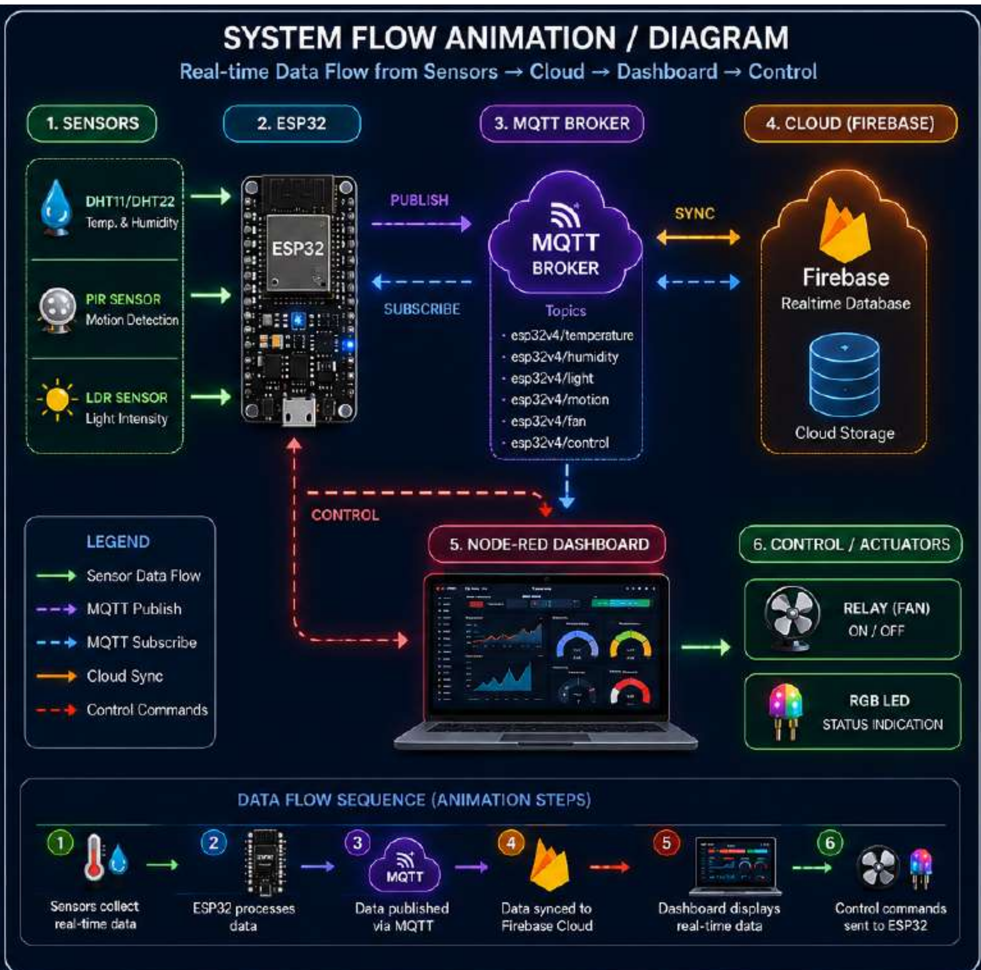
- Start / End
- Initialization
- Process / Action
- Decision
- Data / Cloud
- External System
- → Flow Direction
- ---→ MQTT Publish
- ---→ MQTT Subscribe

SYSTEM FEATURES

- Real-time Monitoring (Temp, Humidity, Light, Motion)
- Automatic Fan Control
- Smart RGB LED Status Indication
- Manual & Automatic Modes
- MQTT Based Communication
- Node-RED Dashboard
- Cloud Storage with Firebase

SMART ROOM V4
SPECIFICATIONS

ITEM	DETAILS
Voltage	5V DC (via USB or External Power Supply)
WiFi	2.4GHz 802.11 b/g/n (ESP32 Built-in WiFi)
Sensors	• DHT11 / DHT22 (Temperature & Humidity) • PIR Sensor (Motion Detection) • LDR Sensor (Light Intensity)
MQTT Broker	broker.emqx.io (Public MQTT Broker) Port: 1883 | Protocol: MQTT v3.1.1
Database	Firebase Realtime Database Cloud Sync (JSON Data)
Response Time	~200 – 500 ms (End-to-End from Sensor to Dashboard)
Microcontroller	ESP32 Dev Module Dual Core 240MHz, 520KB SRAM
Display	0.96" OLED Display (I2C) 128x64 Pixels
Actuators	Relay Module (Fan Control) RGB LED (Status Indication)
Communication	WiFi + MQTT (Publish / Subscribe) JSON Data Format
Dashboard	Node-RED Dashboard (Real-time UI) Accessible via Web Browser
Cloud Sync	Realtime Data Push to Firebase MQTT Bi-directional Communication



SYSTEM BENEFITS

- **Real-time Monitoring** – Live room data updates
- **Smart Automation** – Auto fan & lights control
- **Cloud Connected** – Access anywhere
- **Low Latency** – Fast response (~200–500ms)
- **Scalable** – Easy to expand

Advanced IoT Smart Room Automation System

ESP32 • MQTT • Node-RED • Firebase Realtime Database

Prepared by: Ahmad Azroun

Renewable Energy Manager | IoT & Smart Energy Systems Developer

DEUTSCHE DOKUMENTATION

Projektübersicht:
Smart Room V4 ist ein professionelles IoT-Automatisierungssystem basierend auf ESP32, MQTT, Node-RED und Firebase Realtime Database. Das System wurde entwickelt, um Raumdaten intelligent zu überwachen und automatisch auf Umgebungsänderungen zu reagieren.

Projektfunktionen:
• Temperatur- und Luftfeuchtigkeitsüberwachung
• Bewegungserkennung mit PIR-Sensor
• Lichtintensitätsmessung mit LDR
• Automatische Lüftersteuerung
• RGB-LED Statusanzeigen
• Echtzeit-Dashboard mit Node-RED
• Cloud-Datenspeicherung über Firebase
• MQTT-basierte Kommunikation
• Manueller und automatischer Betriebsmodus

Arbeitsprinzip:
Die Sensoren senden Daten an den ESP32. Anschließend verarbeitet der ESP32 die Werte und sendet die Daten über MQTT an Node-RED und Firebase. Das System steuert automatisch Lüfter und RGB-Beleuchtung basierend auf Temperatur, Licht und Bewegungserkennung.

ENGLISH DOCUMENTATION

Project Overview:
Smart Room V4 is a professional IoT automation system based on ESP32, MQTT, Node-RED, and Firebase Realtime Database. The system was designed to intelligently monitor room conditions and automatically react to environmental changes.

Main Features:
• Temperature and humidity monitoring
• PIR motion detection
• Light intensity measurement using LDR
• Automatic fan control system
• RGB LED smart status indication
• Real-time dashboard using Node-RED
• Cloud storage using Firebase
• MQTT-based communication
• Manual and automatic operating modes

Working Principle:
Sensors continuously send realtime data to the ESP32 controller. The ESP32 processes the values and publishes them through MQTT to Node-RED and Firebase. The system automatically controls the cooling fan and RGB lighting based on motion, temperature, and room light conditions.

Component	Purpose
ESP32	Main controller and Wi-Fi communication
DHT11/DHT22	Temperature and humidity sensing
LDR Sensor	Light intensity monitoring
PIR Sensor	Motion detection
RGB LED	Visual room status indication
Relay Module	Fan switching control
Cooling Fan	Automatic room cooling
OLED Display	Realtime local display
Node-RED	Realtime dashboard and visualization
Firebase	Cloud realtime database storage

Architecture Diagram

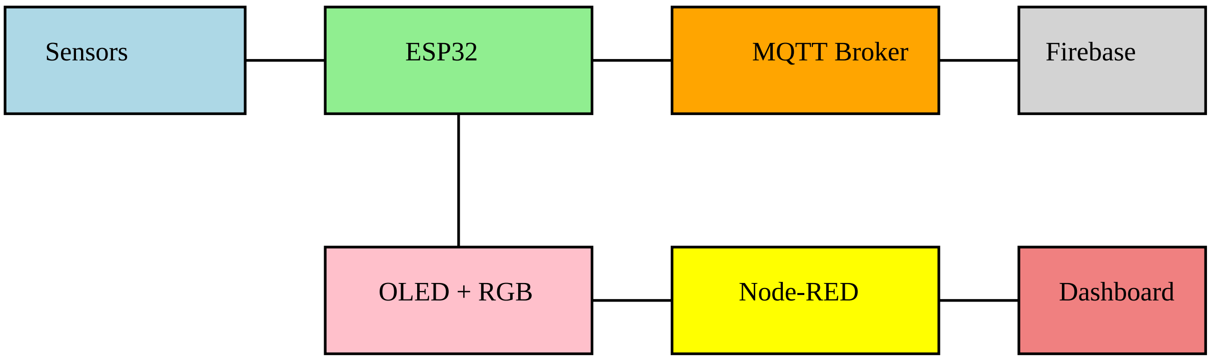
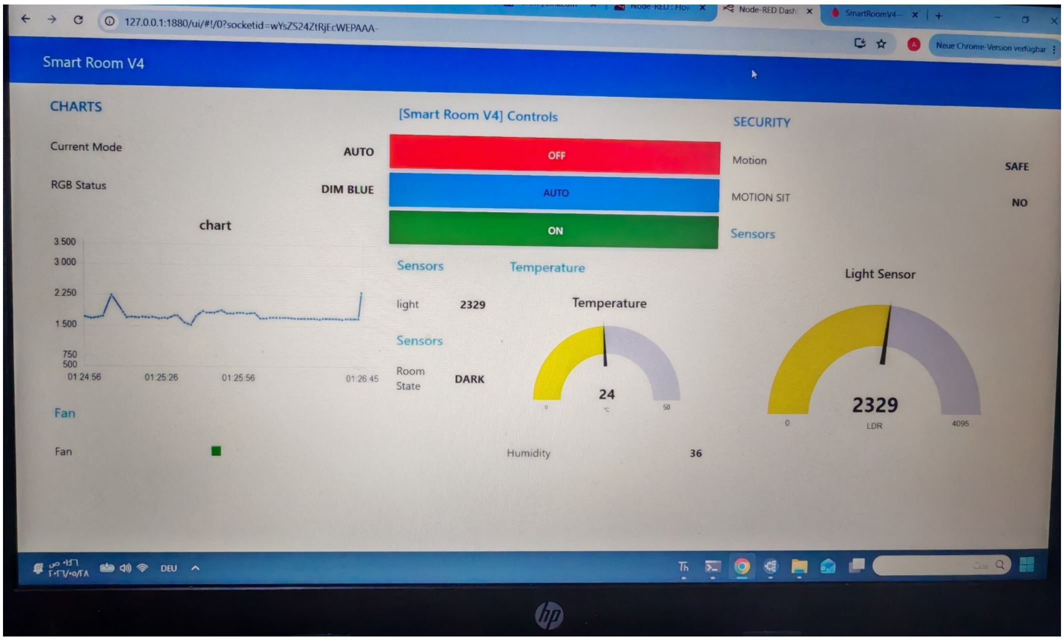
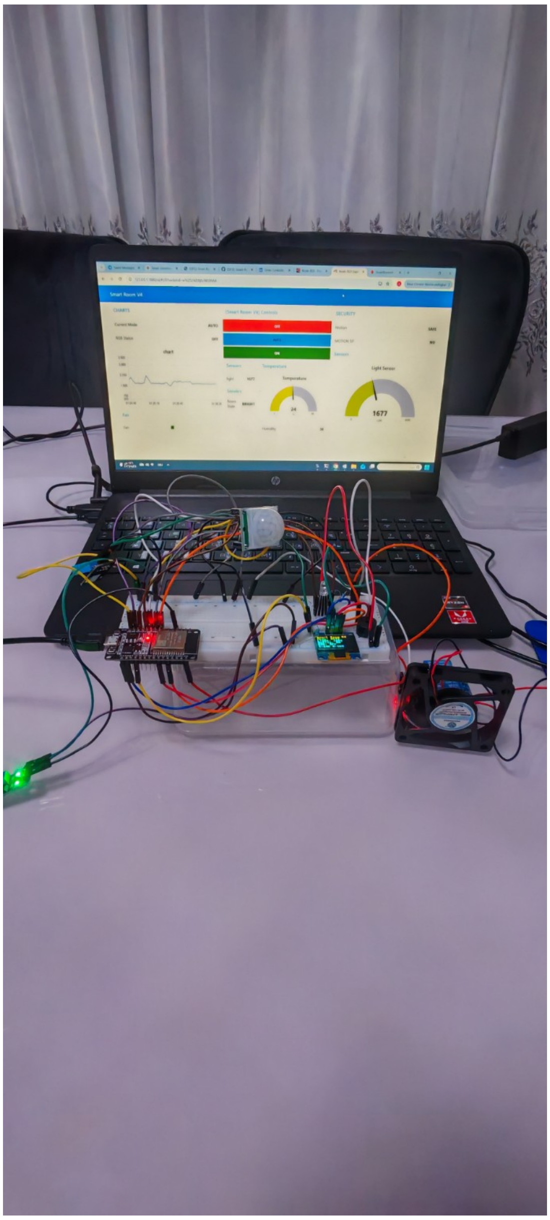


Figure 1



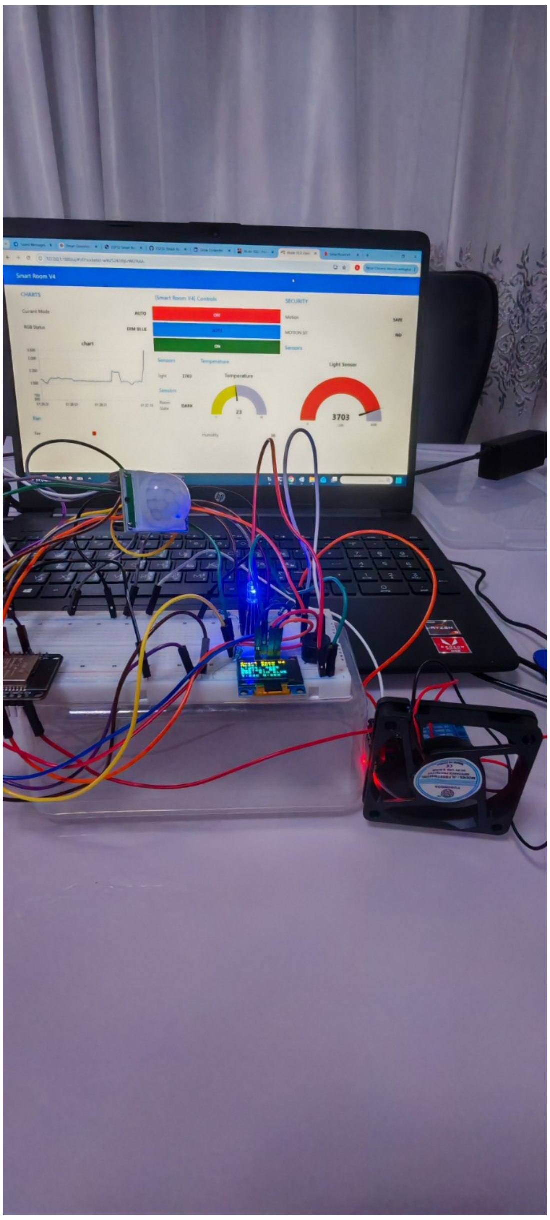
Node-RED dashboard showing realtime room monitoring and smart controls.

Figure 2



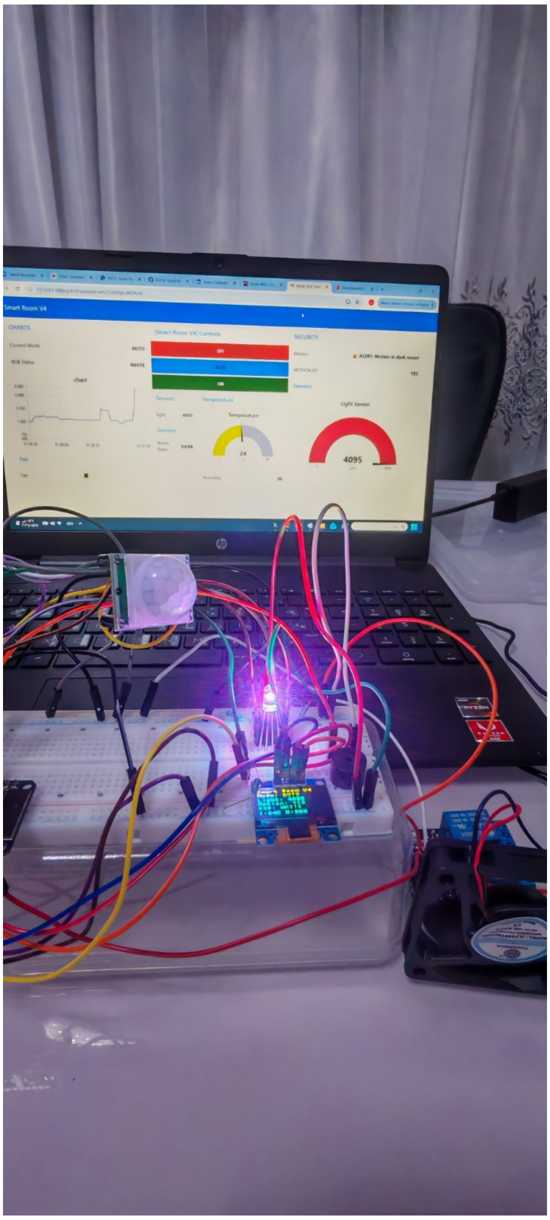
Complete Smart Room V4 hardware setup including ESP32, sensors, RGB LED, OLED, and cooling fan.

Figure 3



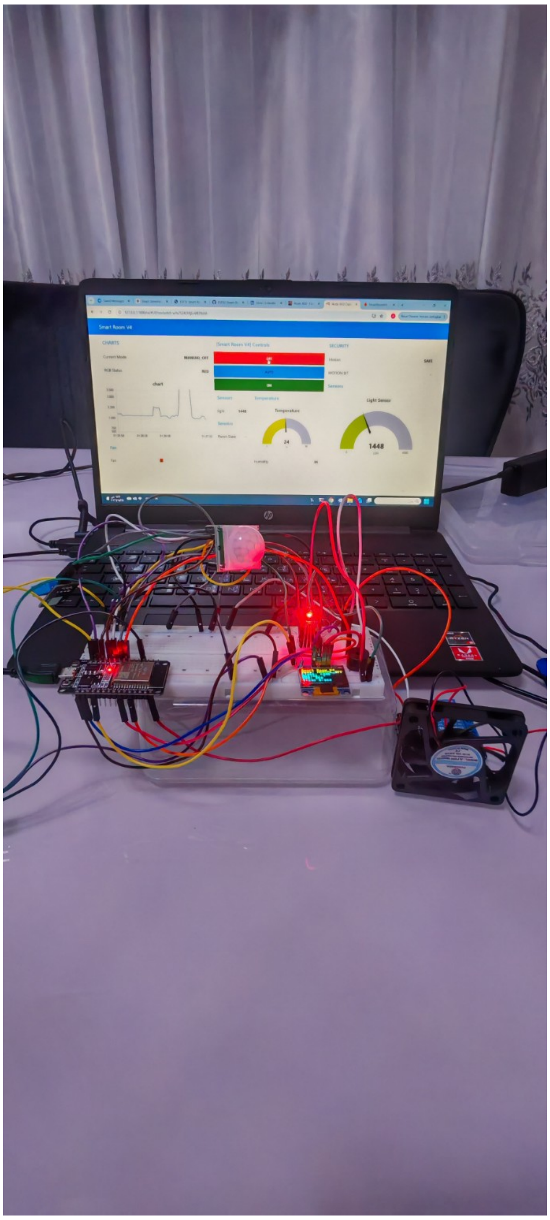
Automatic dark mode with blue RGB LED indicating low-light environment.

Figure 4



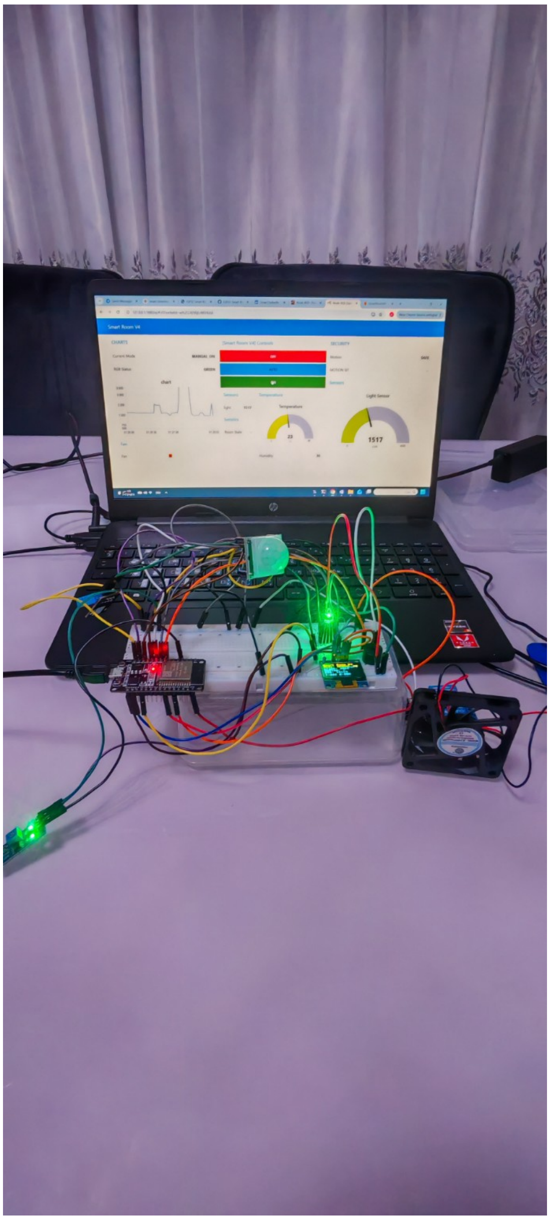
Motion detection mode activating white RGB LED alert mode.

Figure 5



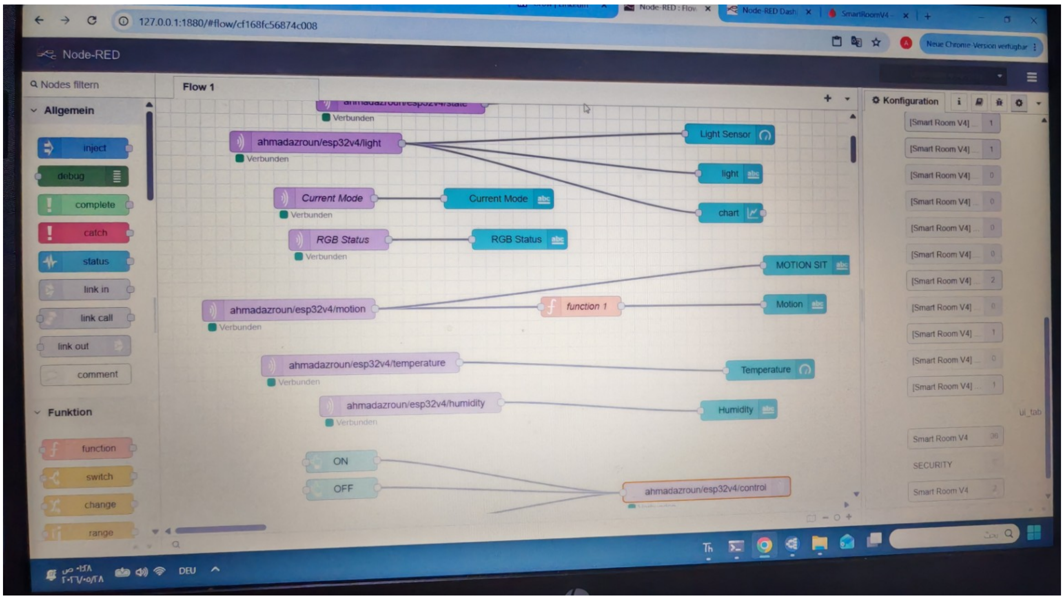
Manual OFF mode using red RGB LED indication.

Figure 6



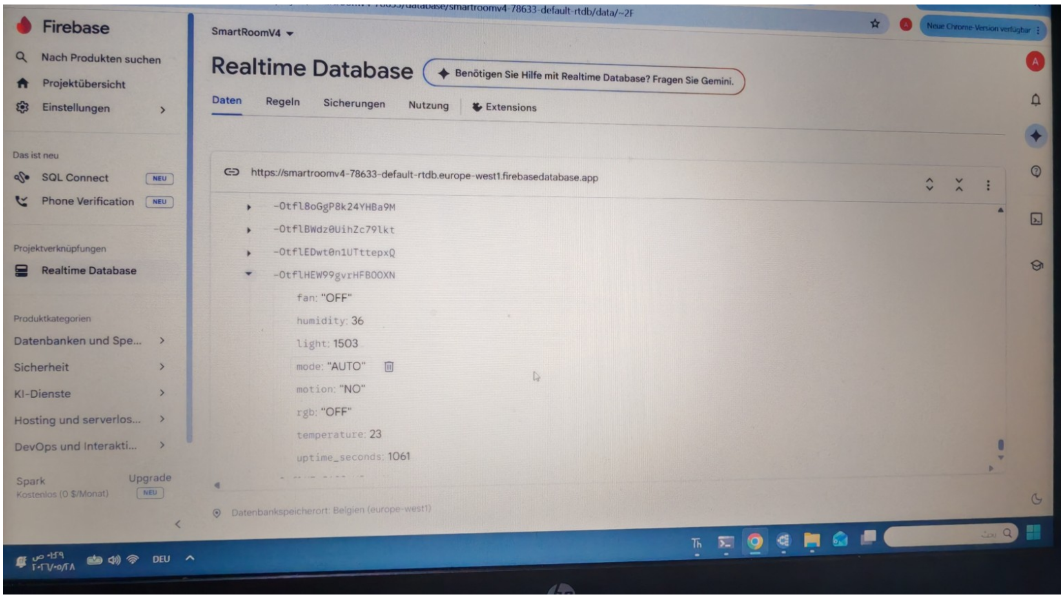
Manual ON mode using green RGB LED indication.

Figure 7



Node-RED MQTT flow configuration and sensor routing.

Figure 8



Firebase Realtime Database storing live IoT data values.

Practical Applications

• Smart home automation systems
• Intelligent energy-saving solutions
• Realtime environmental monitoring
• Security and motion detection systems
• Industrial automation monitoring
• Cloud-connected IoT applications
• Smart office and classroom automation
• Remote monitoring dashboards

Conclusion

Smart Room V4 successfully combines embedded systems engineering, IoT networking, realtime cloud communication, dashboard visualization, and intelligent automation into one integrated smart environment solution. The project demonstrates advanced practical implementation of ESP32, MQTT communication, Firebase cloud storage, and Node-RED dashboard integration in a professional IoT automation platform.